Advanced Engineering Report: Structural Integrity & Stress Analysis

Electric Wheelchair Chassis for High-Load Applications

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1. Abstract

This technical report delineates a comprehensive Finite Element Analysis (FEA) performed on a custom-engineered electric wheelchair frame. The primary objective is to validate the structural stability under a maximum static payload of 120 kg. The analysis employs the Von Mises yield criterion to assess potential failure points and ensure a robust Factor of Safety (FoS) consistent with international medical device standards.

2. Design Specifications & Geometrical Parameters

The following table summarizes the critical dimensions of the prototype as measured from the physical assembly:

Parameter	Metric Value	Unit
Total Height	88	cm
Ground to Armrest Height	67	cm
Internal Seat Width	48	cm
External Total Width	59	cm
Rear Drive Wheel Diameter	30	cm

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Matters Need Attention

3. Material Characterization & Load Modeling

The chassis is constructed from high-strength structural alloy (Aluminum 6061-T6) to optimize the strength-to-weight ratio.

3.1 Loading Conditions:

- **Occupant Load (P_occ):** 1177.2 N (Equivalent to 120 kg mass).
- **Drivetrain Component Load (P_sys):** 200 N (Estimated combined weight of batteries and motors).
- **Loading Strategy:** Applied as a distributed remote load on the seat support rails.

3.2 Theoretical Stress Estimation:

Using Euler-Bernoulli beam theory for the primary horizontal supports:

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Maximum Bending Stress (σ_max) = (M * y) / I
Where M is the bending moment, I is the moment of inertia, and y is
the distance from the neutral axis.
Calculated Theoretical Safety Factor: ~3.05
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4. Finite Element Analysis (FEA) Results

Simulation conducted via ANSYS Mechanical to determine stress concentration zones.

4.1 Von Mises Stress Distribution:

The maximum equivalent stress was identified at the welded junctions of the rear motor brackets. The peak value reached 86.2 MPa, which remains well below the yield strength of 275 MPa.

4.2 Structural Factor of Safety (FoS):

The minimum Factor of Safety recorded was 3.19. This confirms that the design can handle over 3 times the specified 120 kg load before reaching plastic deformation.

(Green Safety Factor)

5. Conclusion & Technical Recommendations

The structural analysis verifies that the wheelchair frame is highly resilient. To further enhance the design, the following recommendations are proposed:

1. **Weldment Inspection:** Critical nodes should undergo ultrasonic testing to ensure zero porosity in high-stress zones.
2. **Dynamic Dampening:** Integration of rubber bushings at motor mountings to reduce cyclic fatigue loading.
3. **Mass Optimization:** Potential for wall-thickness reduction in non-critical vertical members to reduce overall weight without compromising FoS.

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